Frédéric **Andrian**

Software Engineer

(+33) 769875327 | fredericj.andrian@gmail.com | AndriantsehenoFrederic | FredericAndriantseheno

Profile

Recent graduate of ENSC, Bordeaux INP, specializing in artificial intelligence and cognitive science. I recently completed my final-year internship at INRIA Bordeaux and am available as soon as possible. At INRIA, I developed an end-to-end AI application combining a multi-step agent, Python/FastAPI services, a RAG pipeline, MongoDB and a React/Next.js interface. I also designed its prompts and tools, evaluated several LLMs and contributed to a scientific article. I want to continue developing useful and robust AI solutions.

Experience

INRIA — Bordeaux, France

AI Research Engineering Intern — Mar. 2026 – Sept. 2026

- Independently designed and built a working application to help researchers find scientific papers, with guidance from my supervisors.
- Developed a Python/FastAPI backend with REST APIs documented using OpenAPI and MongoDB persistence for documents and metadata.
- Automated online scientific-paper metadata collection in Python and delivered cron task results through webhooks.
- Built a React/Next.js interface for interacting with the agent and its tools, including user selection of messages to include in the model context.
- Integrated the OpenAI API into a custom multi-step agent without an orchestration framework, implementing tool calling, state and memory management.
- Containerized components with Docker and set up a GitHub CI/CD pipeline.
- Managed a web application using models executed locally with LM Studio on a Mac Studio, including model installation and configuration.
- Implemented a RAG pipeline with chunking and embeddings, combining MongoDB for documents and metadata with ChromaDB for vector storage and semantic retrieval.
- Defined experimental protocols to compare LLM quality and latency, using vLLM on HPC with NVIDIA GPUs and Slurm. Currently writing a scientific article presenting the results.

Karunya University — Coimbatore, India

Reinforcement Learning Research Intern — June 2025 – Sept. 2025

- Developed a reinforcement learning model and its training environment for autonomous navigation of a hospital robot.
- Deployed and optimized the software and model on a Raspberry Pi for real-world testing.

Academic Projects

University Management REST API — Academic project

C#, ASP.NET Core, Entity Framework Core, SQLite

- Developed a REST API in C# and ASP.NET Core for university data management.
- Used DTOs to structure the data exchanged through the API.
- Implemented data persistence with Entity Framework Core and a SQLite SQL database.
- Documented the API using Swagger/OpenAPI.

Music Quiz Web Application — Academic project

React, Spotify API

- Built a React web application integrating the Spotify API to create music quizzes, including blind tests.
- Deployed the application to make it accessible to users.

Skills

Development	Python, JavaScript, C#, .NET, Java, FastAPI, React, Next.js, Node.js, REST APIs, OpenAPI
Generative AI and agents	LLMs, vLLM, OpenAI API, LangChain, prompts, multi-step agents, RAG, embeddings, ChromaDB
Data and Machine Learning	Pandas, scikit-learn, SQL, MongoDB, PyTorch, TensorFlow, Deep Learning, RL
LLM evaluation	Experimental protocols, quantization, latency, TTFT, quality, context length
Software engineering tools	Linux, Docker, Git/GitHub, CI/CD, Slurm, HPC environments
Languages	French (native), English (fluent, IELTS 7.5), Spanish (basic)

Education

ENSC, Bordeaux INP — Bordeaux, France

Engineering Degree, MSc-equivalent, in Artificial Intelligence and Cognitive Science — 2023 – 2026

- Coursework in computer science, software engineering, AI and cognitive science, including human-computer interaction, ergonomics and user experience, with a specialization in LLMs, NLP and deep learning.